3

PROBLEM DEFINITION AND NEED IDENTIFICATION

3.1 INTRODUCTION

The engineering design process has been depicted as a stream of potential designs for a new product that will fit the needs of a targeted group of consumers. The stream is channeled through a pipeline of narrowing diameter with filters at key junctions that screen out less valuable candidate designs, as shown in Fig. 2.1. At the end of the pipeline, a nearly ideal single design (or a very small set of designs) emerges. The filters represent key decision points in the design evaluation process where candidate designs are evaluated by a panel of reviewers overseeing product development for the business unit. Candidate designs are rejected when they fail to meet one or more of the engineering or business objectives of the unit.

This is an optimistic view of designing. Candidate design concepts are not stored and waiting to be released like ice cubes dispensed one glassful at a time from a port in a refrigerator door. Design is a much more complex activity that requires intense focus at the very beginning to determine the full and complete description of what the final product will do for a particular customer base with a set of specific needs. The design process only proceeds into concept generation once the product is so well described that it has met with the approval of groups of technical and business discipline specialists and managers. These review groups include the R&D division of the corporation and may also include employees anywhere in the company, as well as customers and key suppliers. New product ideas must be checked for their fit with the technology and product market strategies of the company, and their requirement for resources. A senior management team will review competing new product development plans championed by different product managers to select those in which to invest resources. The issues involved in planning for the design of a new product are discussed in various sections of Chapter 2 namely: Product and Process Cycles; Markets and Marketing, and Technological Innovation. Certain decisions about the PDP are made even before the process begins. These sections point out certain types

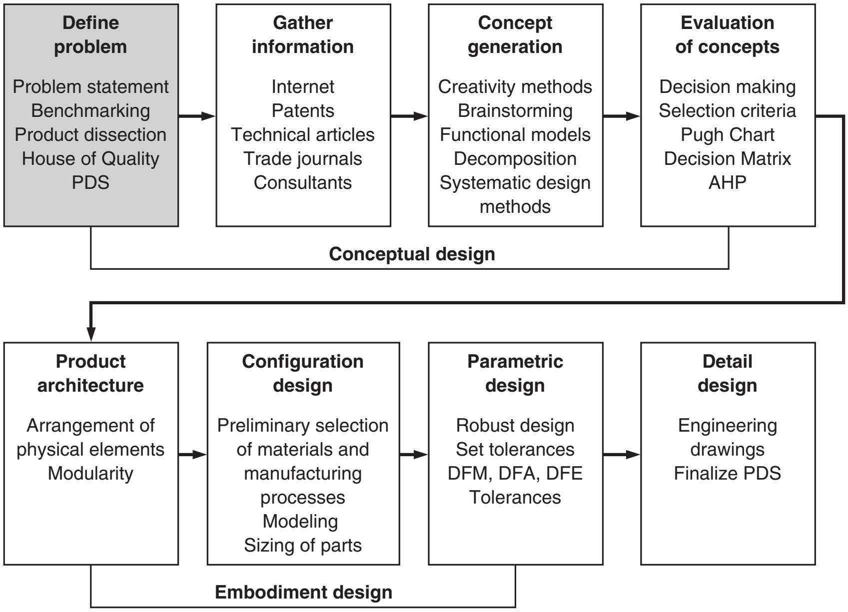

FIGURE 3.1
The product development process showing problem definition as the start of the conceptual design process.

of development work and decision making that must be completed before the design problem definition starts.

Product development begins by determining what the needs are that a product must meet. Problem definition is the most important of the steps in the PDP (Fig. 3.1). As discussed in Sec. 1.3, understanding any problem thoroughly is crucial to reaching an outstanding solution. This axiom holds for all kinds of problem solving, whether it be math problems, production problems, or design problems. In product design the ultimate test of a solution is meeting management's goal in the marketplace, so it is vital to work hard to understand and provide what it is that the customer wants. Fortunately, the product development process introduced in Chap. 2 is a structured methodology that focuses specifically on creating products that will succeed in the marketplace.

This chapter emphasizes the customer satisfaction aspect of problem definition, an approach not always taken in engineering design. This view turns the design problem definition process into the identification of what outcome the customer or end user of the product wants to achieve. *Therefore, in product development, the problem definition process is mainly the need identification step.* The need identification methods in this chapter draw heavily on processes introduced and proven effective by the total quality management (TQM) movement. TQM emphasizes customer satisfaction. The TQM tool of *quality function deployment* (QFD) will be introduced. QFD is a process devised to identify the *voice of the customer* and channel it through the entire product development process. The most popular step of QFD, producing the House of Quality

(HOQ), is presented here in detail. The chapter ends by proposing an outline of the *product design specification* (PDS), which serves as the governing document for the product design. A design team must generate a starting PDS at this point in the design process to guide its design generation. However, the PDS is an evolving document that will not be finalized until the detail design phase of the PDP process.

3.2 IDENTIFYING CUSTOMER NEEDS

Increasing worldwide competitiveness creates a need for greater focus on the customer's wishes. Engineers and businesspeople are seeking answers to such questions as: Who are my customers? What does the customer want? How can the product satisfy the customer while generating a profit?

Webster defines a customer as "one that purchases a product or service." This is the definition of the customer that most people have in mind, the *end user.* These are the people or organizations that buy what the company sells because they are going to be using the product. However, engineers performing product development must broaden their definition of *customer* to be most effective.

From a total quality management viewpoint, the definition of *customer* can be broadened to "anyone who receives or uses what an individual or organization provides." This includes the end users who are also making their own purchasing decisions. However, not all customers who make purchasing decisions are end users. Clearly the parent who is purchasing action figures, clothes, school supplies, and even breakfast cereal for his or her children is the not the end user but still has critical input for product development. Large retail customers who control distribution to a majority of end users also have increasing influence. In the do-it-yourself tool market, Home Depot and Lowes act as customers but they are not end users. Therefore, both customers *and* those who influence them must be consulted to identify needs the new product must satisfy. This strategy of focusing on customers' needs can be very powerful, as demonstrated by the impact it has had on Advanced Micro Devices (see the following article).

The needs of customers outside of the company are important to the development of the product design specifications for new or improved products. A second set of critical constituents are the internal customers, such as a company's own corporate management, manufacturing personnel, the sales staff, and field service personnel must be considered. For example, the design engineer who receives information on the properties of three potential materials for his or her design is an *internal customer* of the materials specialist.

The product under development defines the range of customers that a design team must consider. Remember that the term *customer* implies that the person is engaging in more than just a one-time transaction. Every great company strives to convert each new buyer into a customer for life by delivering quality products and services. A customer base is not necessarily captured by a fixed demographic range. Marketing professionals are attuned to changes in customer bases that will lead to new definitions of markets for existing product improvements and new target markets for product innovations.

3

As Intel Slips, Smaller AMD Makes Strides

Since the early 1990s, computer makers' profits have paled compared with those of two suppliers—Microsoft Corp., for software, and Intel Corp., for chips that provide the calculating power in personal computers.

But the hardware half of that picture is suddenly looking fuzzy. That is partly because Hector Ruiz, chief executive of lntel rival Advanced Micro Devices Inc., has pushed his company to treat customers like partners. AMD, once an unreliable also-ran in the microprocessor market, has exploited computer makers' suggestions to gain advantages that Intel is struggling to match. AMD's technology is even starting to find converts among corporate computer buyers who long favored the "Intel Inside" brand . . . .

The chip makers' contrasting fortunes became glaringly obvious this week. Intel, though it has more than six times AMD's revenue, posted lower sales and said conditions would get worse in the second period. Its closely watched gross profit margin was 55.1%—below its prediction in January of 59%—and the company said it could sink to 49% in the current period. At AMD, meanwhile, microprocessor sales surged and its profit margin topped Intel's, at 58.5%—a rarity in the companies' 25-year rivalry . . . .

AMD's Aladdin is Mr. Ruiz, a Mexican-born engineer who worked for 22 years at Motorola Inc. before joining AMD in 2000. He succeeded Jerry Sanders, one of Silicon Valley's most flamboyant executives, and brought a more understated, methodical style to the company. Mr. Ruiz, 60 years old, replaced most senior managers, improved manufacturing efficiency and, most recently, spun off a memory-chip unit that was holding down profit. . .

Mr. Ruiz has stepped up what he calls "customer-centric" innovation—taking customers' suggestions that have led AMD to scoop Intel with some attractive features. In other cases, AMD has heeded requests to wait for lower prices before adopting new technology. "The reason AMD is being so practical is they can't afford to do it any other way," says John Fowler, an executive vice president at AMD customer Sun Microsystems Inc.

AMD's strategy is tailored to computer makers' desire for a choice of suppliers, to help them differentiate products and pit one supplier against another to lower prices. H-P is an example. Nearly a decade ago, H-P first selected AMD for some consumer PCs after the smaller vendor agreed to tweak its technology to help H-P develop a system that could sell for less than $1,000. Intel declined, H-P executives say . . . .

Customer suggestions have been particularly important in servers, which Mr. Ruiz targeted first to impress the most demanding technology buyers at corporations. AMD, for example, built one model of its Opteron chips specifically in response to a Sun suggestion, Mr. Fowler says.

The company has also picked the brains of boutique PC makers. Rahul Sood, president and chief technology officer of Voodoo PC, recalls meetings where AMD officials agreed to change chips' features and names at the request of his company or other makers of machines for gamers. "One of the products that we suggested to them is going to become a reality, which is unbelievable," Mr. Sood says.

From *The Wall Street Journal*, April 21, 2006.

3.2.1 Preliminary Research on Customers Needs

In a large company, the research on customer needs for a particular product or for the development of a new product is done using a number of formal methods and by different business units. The initial work may be done by a marketing department specialist or a team made up of marketing and design professionals (see Sec. 2.5). The natural focus of marketing specialists is the buyer of the product and similar products. Designers focus on needs that are unmet in the marketplace, products that are similar to the proposed product, historical ways of meeting the need and technological approaches to engineering similar products of the type under consideration. Clearly, information gathering is critical for this stage of design. Chapter 5 outlines sources and search strategies for finding published information on existing designs. Design teams will also need to gather information directly from potential customers.

One way to begin to understand needs of the targeted customers is for the development team to use their own experience and research to date. The team can begin to identify the needs that current products in their area of interest do not meet and those that an ideal new product should meet. In fact, there's no better group of people to start articulating unmet needs than members of a product development team who also happen to be end users of what they are designing.

Brainstorming is a natural idea generation tool that can be used at this point in the process. Brainstorming will be covered in more detail in Chap. 4. It is such a familiar process that a brief example of how brainstorming can be carried out to provide insight into customer needs is given here.

EXAMPLE 3.1

A student design team[1] selected the familiar "jewel case" that protects compact discs in storage as a product needing improvement. As a first step, the team brainstormed to develop ideas for possible improvements to the CD case (Table. 3.1). The following ideas were generated in response to the question: What improvements to the current CD case would customers want?

1. Case more resistant to cracking
2. Easier to open
3. Add color
4. Better waterproofing
5. Make it lighter
6. More scratch-resistant
7. Easier extraction of CD from the circular fastener
8. Streamlined look
9. Case should fit the hand better
10. Easier to take out leaflet describing the CD
11. Use recyclable plastic
12. Make interlocking cases so they stack on top of each other without slipping

1. The original design task was developed and performed by a team composed of business and engineering students at the University of Maryland. Team members were Barry Chen, Charles Goldman, Annie Kim, Vikas Mahajan, Kathy Naftalin, Max Rubin, and Adam Waxman. The results of their study have been augmented and modified significantly by the authors.

3

TABLE 3.1
Affinity Diagram Created from Brainstormed CD Case Design Improvements

Stronger	Aesthetics	Opening and Extracting	Environment	Other
1	3	2	4	12
6	5	7	11	
14	8	10		
	9	13		

13. Better locking case
14. Hinge that doesn't come apart

Next, the ideas for improvement were grouped into common areas by using an *affinity diagram* (see Chap. 4). A good way to achieve this is to write each of the ideas on a Post-it note and place them randomly on a wall. The team then examines the ideas and arranges them into columns of logical groups. After grouping, the team determines a heading for the column and places that heading at the top of the column. The team created an affinity diagram for their improvement ideas, and it is shown in Table 3.1.

The five product improvement categories appearing in Table 3.1 emerged from the within-team brainstorming session. This information helps to focus the team's design scope. It also aids the team in determining areas of particular interest for more research from direct interaction with customers and from the team's own testing processes.

3.2.2 Gathering Information from Customers

It is the customer's desires that ordinarily drive the development of the product, not the engineer's vision of what the customer should want. (An exception to this rule is the case of technology driving innovative products that customers have never seen before, Sec. 2.6.4.) Information on the customer's needs is obtained through a variety of channels[2]:

- *Interviews with customers:* Active marketing and sales forces should be continuously meeting with current and potential customers. Some corporations have account teams whose responsibility is to visit key customer accounts to probe for problem areas and to cultivate and maintain friendly contact. They report information on current product strengths and weaknesses that will be helpful in product upgrades. An even better approach is for the design team to interview single customers in the service environment where the product will be used. Key questions to ask are: What do you like or dislike about this product? What factors do you consider when purchasing this product? What improvements would you make to this product?

2. K. T. Ulrich and S. D. Eppinger, *Product Design and Development,* 3rd ed., McGraw-Hill, New York, 2004.

- *Focus groups:* A focus group is a moderated discussion with 6 to 12 customers or targeted customers of a product. The moderator is a facilitator who uses prepared questions to guide the discussion about the merits and disadvantages of the product. Often the focus group occurs in a room with a one-way window that provides for videotaping of the discussion. In both the interviews and the focus groups it is important to record the customer's response in his or her own words. All interpretation is withheld until the analysis of results. A trained moderator will follow up on any surprise answers in an attempt to uncover implicit needs and latent needs of which the customer is not consciously aware.
- *Customer complaints:* A sure way to learn about needs for product improvement is from customer complaints. These may be recorded by communications (by telephone, letter, or email) to a customer information department, service center or warranty department, or a return center at a larger retail outlet. Third party Internet websites can be another source of customer input on customer satisfaction with a product. Purchase sites often include customer rating information. Savvy marketing departments monitor these sites for information on their products and competing products.
- *Warranty data:* Product service centers and warranty departments are a rich and important source of data on the quality of an existing product. Statistics on warranty claims can pinpoint design defects. However, gross return numbers can be misleading. Some merchandise is returned with no apparent defect. This reflects customer dissatisfaction with paying for things, not with the product.
- *Customer surveys:* A written questionnaire is best used for gaining opinions about the redesign of existing products or new products that are well understood by the public. (Innovative new products are better explored with interviews or focus groups.) Other common reasons for conducting a survey are to identify or prioritize problems and to assess whether an implemented solution to a problem was successful. A survey can be done by mail, e-mail, telephone, or in person.

The creation of customer surveys is now presented in more detail.

Constructing a Survey Instrument

Regardless of the method used to gain information from customers, considerable thought needs to go into developing the survey instrument[3]. Creating an effective survey requires the following steps. A sample survey used for the CD jewel case of Example 3.1 is shown in Figure 3.2.

1. Determine the survey purpose. Write a short paragraph stating the purpose of the survey and what will be done with the results. Be clear about who will use the results.
2. Determine the type of data-collection method to be used. Surveys and closely scripted interviews are effective for compiling quantitative statistics. Focus groups or free-form interviews are useful for collecting qualitative information from current and targeted customers.

---

3. P. Slaat and D. A. Dillman, *How to Conduct Your Own Survey,* Wiley, New York, 1994 and http://www.surveysystem.com/sdesign.htm, accessed July 6, 2006.

3

Compact Disc Case
Product Improvement Survey

A group of students in ENES 190 is attempting to improve the design and usefulness of the standard storage case for compact discs. Please take 10 minutes to fill out this customer survey and return it to the student marketer.

Please indicate the level of importance you attach to the following aspects of a CD case.
1 = low importance 5 = high importance

1. A more crack-resistant case	1	2	3	4	5
2. A more scratch-resistant case	1	2	3	4	5
3. A hinge that doesn't come apart	1	2	3	4	5
4. A more colorful case	1	2	3	4	5
5. A lighter case	1	2	3	4	5
6. A streamlined look (aerodynamically sleek)	1	2	3	4	5
7. A case that fits your hand better	1	2	3	4	5
8. Easier opening CD case	1	2	3	4	5
9. Easier extraction of the CD from the circular fastener	1	2	3	4	5
10. Easier to take out leaflet describing contents of the CD	1	2	3	4	5
11. A more secure locking case	1	2	3	4	5
12. A waterproof case	1	2	3	4	5
13. Make the case from recyclable plastic	1	2	3	4	5
14. Make it so cases interlock so they stack on each other without slipping	1	2	3	4	5

Please list any other improvement features you would like to see in a CD case. ____________________

____

____

Would you be willing to pay more for a CD if the improvements you value with a 5 or 4 rating are available on the market? yes no

If you answered yes to the previous question, how much more would you be willing to pay? ____________

How many CD's do you own (approximately)? ____________

FIGURE 3.2
Customer survey for the compact disc case.

3. Identify what specific information is needed. Each question should have a clear purpose for eliciting responses to inform on specific issues. You should have no more questions than the absolute minimum.
4. Design the questions. Each question should be unbiased, unambiguous, clear, and brief. There are three categories of questions: (1) attitude questions—how the customers feel or think about something; (2) knowledge questions—questions asked to determine whether the customer knows the specifics about a product or service; and (3) behavior questions—usually contain phrases like "how often," "how much," or "when." Some general rules to follow in writing questions are:
   - Do not use jargon or sophisticated vocabulary.
   - Focus very precisely. Every question should focus directly on one specific topic.
   - Use simple sentences. Two or more simple sentences are preferable to one compound sentence.
   - Do not lead the customer toward the answer you want.

- Avoid questions with double negatives because they may create misunderstanding.
- In any list of options given to the respondents, include the choice of “Other” with a space for a write-in answer.
- Always include one open-ended question. Open-ended questions can reveal insights and nuances and tell you things you would never think to ask.
- The number of questions should be such that they can be answered in about 15 (but no more than 30) minutes.
- Design the survey form so that tabulating and analyzing data will be easy.
- Be sure to include instructions for completing and returning it.

Questions can have the following types of answers:

- Yes—no—don’t know
- A Likert-type, rating scale made up of an odd number of rating responses, e.g., strongly disagree—mildly disagree—neutral—mildly agree—strongly agree. On a 1–5 scale such as this, always set up the numerical scale so that a high number means a good answer. The question must be posed so that the rating scale makes sense.
- Rank order—list in descending order of preference
- Unordered choices—choose (b) over (d) or (b) from a, b, c, d, e.

Select the type of answer option that will elicit response in the most revealing format without overtaxing the respondent.

5. Arrange the order of questions so that they provide context to what you are trying to learn from the customer. Group the questions by topic, and start with easy ones.
6. Pilot the survey. Before distributing the survey to the customer, always pilot it on a smaller sample group and review the reported information. This will tell you whether any of the questions are poorly worded and sometimes misunderstood, whether the rating scales are adequate, and whether the questionnaire is too long.
7. Administer the survey. Key issues in administering the survey are whether the people surveyed constitute a representative sample for fulfilling the purpose of the survey, and what size sample must be used to achieve statistically significant results. Answering these questions requires special expertise and experience. Consultants in the area of marketing should be used for really important situations.

Evaluating Customer Surveys

To evaluate the customer responses, we could calculate the average score for each question, using a 1–5 scale. Those questions scoring highest would represent aspects of the product ranked highest in the minds of the customers. Alternatively, we can take the number of times a feature or attribute of a design is mentioned in the survey and divide by the total number of customers surveyed. For the questionnaire shown in Figure 3.2 , we might use the number of responses to each question rating a feature as either a 4 or a 5. The information gathered from customers using the questionnaire on the CD is summarized in Table 3. 2.

It is worth noting that a response to a questionnaire of this type really measures the need obviousness as opposed to need importance. To get at true need importance, it is

TABLE 3.2
Summary of Responses from Customer Survey for CD Case

Question Number	Number of Responses with 4 or 5 Rating	Relative Frequency
1	70	81.4
2	38	44.2
3	38	44.2
4	17	19.8
5	17	19.8
6	20	23.2
7	18	20.9
8	38	44.2
9	40	46.5
10	43	50.0
11	24	27.9
12	36	41.8
13	39	45.3
14	47	54.6

necessary to conduct face-to-face interviews or focus groups and to record the actual words used by the persons interviewed. These responses need to be studied in depth, a tedious process. Also, it is important to realize that often respondents will omit talking about factors that are very important to them because they seem so obvious. Safety and durability are good examples. It is also possible for an end user to forget to mention a feature of a product that has become standard. An example is a remote control with a television or an icemaker with a refrigerator. Not all end user needs are of equal importance to the design process. This is addressed in Section 3.3.2.

The relative frequency of responses from a survey can be displayed in a bar graph or a Pareto chart (Fig. 3.3). In the bar graph the frequency of responses to each of the questions is plotted in order of the question number. In the Pareto chart the frequency of responses is arranged in decreasing order with the item of highest frequency at the left-hand side of the plot. Only questions with more than a 40 percent response rate have been included. This plot clearly identifies the most important customer requirements—the vital few.

Perusal of the bar charts in Fig. 3.3 and the information in Table 3. 2 suggests that the customer is most concerned with a more crack-resistant case (number 1). Following that come the convenience features of being able to stack the cases in a stable, interlocking way (number 14), making it easier to extract the leaflet (number 10), and making it easier to extract the CD (number 9) from the case. This is useful information, but more can be gained by applying other investigative techniques.

Ethnographic Studies

Surveys can be a powerful means of collecting answers to known questions. However, finding out the complete story about how customers interact with a product is

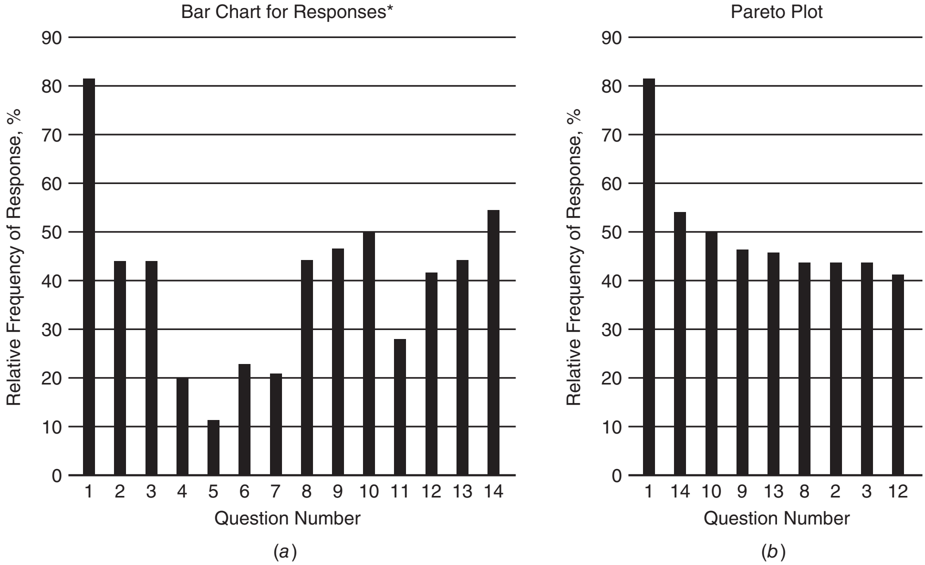

FIGURE 3.3
(a) Frequency of response plotted against question number in a conventional bar graph.
(b) Same data plotted as a Pareto diagram.

*Counts responses for each question that scored either 4 or 5.

often more difficult than asking for answers to a brief survey. Customers are inventive, and much can be discovered from them. A method called ethnographic investigation is valuable to learning about the way people behave in their regular environments.[4] The design team can employ this method to determine how a customer uses (or misuses) a product. Ethnographic study of products involves observing actual end users interacting with the product under typical use conditions. Team members collect photographs, sketches, videos, and interview data during an ethnographic study. The team can further explore product use by playing the roles of typical end users. (A detailed interview with a few end users is more useful than a survey of students acting as end users.)

Ethnography is the process of investigation and documentation of the behavior of a specific group of people under particular conditions. Ethnography entails close observation, even to the point of immersion, in the group being studied while they are experiencing the conditions of interest. This way the observer can get a comprehensive and integrated understanding of the scenario under investigation. It is not unusual for a company to support this type of study by setting up situations that enable members of a product development team to observe end users in their natural work settings. The description of a type of team that Black & Decker created for observing end users and products is given in the following box.

4. Bruce Nussbaum, *Business Week*, June 19, 2006, p.10.

B&D "Swarm" Team Participation Used to Train Engineers

"The engineering development program is a two year development program designed to develop the future engineering talent at Black & Decker. We are looking for strong technical skills, hands-on capabilities and a passion for product design. The purpose is to create technical leaders in all aspects of Black & Decker product development."

"The program consists of four rotating job assignments of four to six months each, plus extensive training courses and seminars."

User-Product Awareness Field Assignment (Swarm Team)
Focus on product knowledge and the customer
Extensive Training at B&D University
Daily contact with end users on a wide range of power tools
Locations nationwide

This is part of the recruiting announcement for Black & Decker's Engineering Development Program for new engineering graduates. (http://www.bdkrotationalprogram.com/edp.asp, Accessed 10/10/06)

3.3 CUSTOMER REQUIREMENTS

Information gathered from customers and research on products from market literature and experimentation contributes to creating a ranked listing of customer needs and wants. These are the needs that form the end user's opinion about the quality of a product. As odd as it may seem, customers may not express all their requirements of a product when they are interviewed. If a feature has become standard on a product (e.g., a remote control on a TV) it is still a need but no longer excites the end users, and they may forget to mention it. To understand how that can happen and how the omissions can be mitigated, it is necessary to reflect on how customers perceive "needs."

From a global viewpoint, we should recognize that there is a hierarchy of human needs that motivate individuals in general.[5]

- *Physiological needs* such as thirst, hunger, sex, sleep, shelter, and exercise. These constitute the basic needs of the body, and until they are satisfied, they remain the prime influence on the individual's behavior.
- *Safety and security needs,* which include protection against danger, deprivation, and threat. When the bodily needs are satisfied, the safety and security needs become dominant.
- *Social needs* for love and esteem by others. These needs include belonging to groups, group identity, and social acceptance.

5. A. H. Maslow, *Psych. Rev.,* vol. 50, pp. 370–396, 1943.

Basic need	Problem situation I	II	III
Food	Hunger	Vitamin deficiency	Food additives
Shelter	Freezing	Cold	Comfort
Work	Availability	Right to work	Work fulfillment

Problem situation	Analysis of problem	Societal perception of need
I	None required	Complete agreement
II	Definition of problem Calculation of cost Setting of priorities	Some disagreement in priorities
III	Analysis of present and future costs Analysis of present and future risks Environmental impact	Strong disagreement on most issues

FIGURE 3.4
A hierarchy of human need situations.

- *Psychological needs* for self-esteem and self-respect and for accomplishment and recognition.
- *Self-fulfillment needs* for the realization of one's full potential through self-development, creativity, and self-expression.

As each need in this hierarchy is satisfied, the emphasis shifts to the next higher need. Our design problem should be related to the basic human needs, some of which may be so obvious that in our modern technological society they are taken for granted. However, within each basic need there is a hierarchy of problem situations.[6] As the type I problem situations are solved, we move to the solution of higher-level problems within each category of basic need. It is characteristic of our advanced affluent society that, as we move toward the solution of type II and III problem situations, the perception of the need by society as a whole becomes less universal. This is illustrated in Fig 3.4. Many current design problems deal with type III situations in which there is strong (type II) societal disagreement over needs and the accompanying goals. The result is protracted delays and increasing costs.

3.3.1 Differing Views of Customer Requirements

From a design team point of view, the customer requirements fit into a broader picture of the PDP requirements, which include product performance, time to market, cost, and quality.

- *Performance* deals with what the design should do when it is completed and in operation. Design teams do not blindly adopt the customer requirements set determined thus far. However, that set is the foundation used by the design team. Other

6. Based on ideas of Prof. K. Almenas, University of Maryland.

factors may include requirements by internal customers (e.g., manufacturing) or large retail distributors.

- The *time* dimension includes all time aspects of the design. Currently, much effort is being given to reducing the PDP cycle time, also known as the time to market, for new products.[7] For many consumer products, the first to market with a great product captures the market (Figure 2.2).
- *Cost* pertains to all monetary aspects of the design. It is a paramount consideration of the design team. When all other customer requirements are roughly equal, cost determines most customers' buying decisions. From the design team's point of view, cost is a result of many design decisions and must often be used to make trade-offs among features and deadlines.
- *Quality* is a complex characteristic with many aspects and definitions. A good definition of quality for the design team is the totality of features and characteristics of a product or service that bear on its ability to satisfy stated or implied needs.

A more inclusive customer requirement than the four listed above is value. Value is the worth of a product or service. It can be expressed by the function provided divided by the cost, or the quality provided divided by the cost. Studies of large, successful companies have shown that the return on investment correlated with high market share and high quality.

Garvin[8] identified the *eight basic dimensions of quality* for a manufactured product. These have become a standard list that design teams use as a guide for completeness of customer requirement data gathered in the PDP.

- *Performance:* The primary operating characteristics of a product. This dimension of quality can be expressed in measurable quantities, and therefore can be ranked objectively.
- *Features:* Those characteristics that supplement a product's basic functions. Features are frequently used to customize or personalize a product to the customer's taste.
- *Reliability:* The probability of a product failing or malfunctioning within a specified time period. See Chap. 13.
- *Durability:* A measure of the amount of use one gets from a product before it breaks down and replacement is preferable to continued repair. Durability is a measure of product life. Durability and reliability are closely related.
- *Serviceability:* Ease and time to repair after breakdown. Other issues are courtesy and competence of repair personnel and cost and ease of repair.
- *Conformance:* The degree to which a product's design and operating characteristics meet both customer expectations and established standards. These standards include industry standards and safety and environmental standards. The dimensions of performance, features, and conformance are interrelated. When competing products have essentially the same performance and many of the same features, customers will tend to expect that all producers of the product will have the same quality dimensions. In other words, customer expectations set the baseline for the product's conformance.

---

7. G. Stalk, Jr., and T. M. Hout, *Competing against Time,* The Free Press, New York, 1990.
8. D. A. Garvin, *Harvard Business Review,* November–December 1987, pp. 101–9.

- *Aesthetics:* How a product looks, feels, sounds, tastes, and smells. The customer response in this dimension is a matter of personal judgment and individual preference. This area of design is chiefly the domain of the *industrial designer,* who is more an artist than an engineer. An important technical issue that affects aesthetics is *ergonomics,* how well the design fits the human user.
- *Perceived quality:* This dimension generally is associated with reputation. Advertising helps to develop this dimension of quality, but it is basically the quality of similar products previously produced by the manufacturer that influences reputation.

The challenge for the design team is to combine all the information gathered about customers' needs for a product and interpret it. The customer data must be filtered into a manageable set of requirements that drive the generation of design concepts. The design team must clearly identify preference levels among the customer requirements before adding in considerations like time to market or the requirements of the company's internal customers.

3.3.2 Classifying Customer Requirements

Not all customer requirements are equal. This essentially means that customer requirements (or their baseline level of Garvin's dimensions for a *quality* product) have different values for different people. The design team must identify those requirements that are most important to the success of the product in its target market and must ensure that those requirements and the needs they meet for the customers are satisfied by the product.

This is a difficult distinction for some design team members to make because the pure engineering viewpoint is to deliver the best possible performance in all product aspects. A Kano diagram is a good tool to visually partition customer requirements into categories that will allow for their prioritization. Kano recognized that there are four levels of customer requirements: (1) expecters, (2) spokens, (3) unspokens, and (4) exciters.[9]

- *Expecters:* These are the basic attributes that one would expect to see in the product, i.e., standard features. Expecters are frequently easy to measure and are used often in benchmarking.
- *Spokens:* These are the specific features that customers say they want in the product. Because the customer defines the product in terms of these attributes, the designer must be willing to provide them to satisfy the customer.
- *Unspokens:* These are product attributes the customer does not generally talk about, but they remain important to him or her. They cannot be ignored. They may be attributes the customer simply forgot to mention or was unwilling to talk about or simply does not realize he or she wants. It takes great skill on the part of the design team to identify the unspoken requirements.
- *Exciters:* Often called *delighters,* these are product features that make the product unique and distinguish it from the competition. Note that the absence of an exciter will not make customers unhappy, since they do not know what is missing.

9. L. Cohen, *Quality Function Deployment: How to Make QFD Work for You*, Addison-Wesley, *Publishing Company,* New York, 1995.

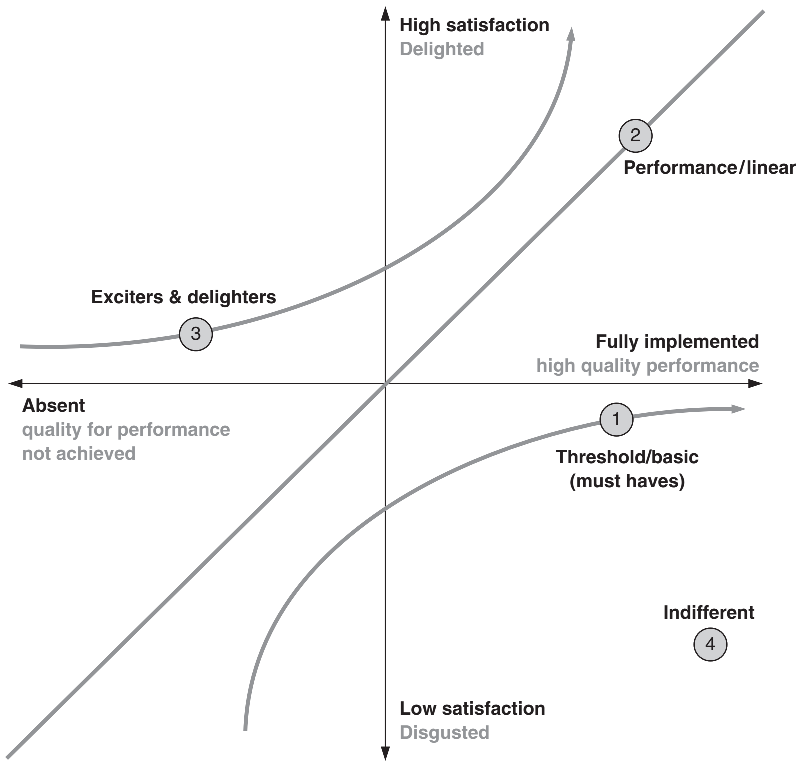

FIGURE 3.5
Kano Diagram

A Kano diagram depicts how expected customer satisfaction (shown on y-axis) can vary with the success of the execution (shown on x-axis) for customer requirements. The success of execution can also be interpreted as product performance. The adequate level of performance is at the zero point on the x-axis. Performance to the right of the y-axis indicates higher quality than required. Performance to the left represents decreasing quality to the point where there is no performance on a requirement.

Figure 3.5 depicts three types of relationships between product performance and customer requirements. Curve 2 is the 45° line that begins in the region of "absent" performance on a requirement and lowest customer satisfaction or "disgust" and progresses to the point of high quality performance and customer delight. Since it is a straight line, it represents customer requirements that are basic to the intended function of the product and will, eventually, result in delight. Customer Requirements (CRs) in the Expecter category are represented on Curve 2. Most Spoken CRs also follow Curve 2.

Curve 1 on Fig. 3.5 begins in the region of existing but less than adequately implemented performance and rises asymptotically to the positive x-axis. Curve 1 will never contribute to positive customer satisfaction. In other words, improving product performance beyond a basic level that contributes to satisfying these CRs will not improve customer perceptions of the quality of the product. However, failing to meet the expected performance will disproportionately decrease quality perceptions. Expecter CRs follow Curve 1. Unspoken CRs that are so expected that customers think they don't have to mention them will also follow Curve 1.

Curve 3 is the mirror image of Curve 1. Any product performance that helps to satisfy these CRs will increase the customer's impression of quality. The improvement in quality rating will increase dramatically as product performance increases. These are the CRs in the Exciter category. The Kano diagram of Fig. 3.5 shows that a design team must be aware of the nature of each CR so that they know which ones are the most important to meet. This understanding of the nature of CRs is necessary for prioritizing design team efforts and making decisions on performance trade-offs.

This partition of customer requirements is hierarchical in a way that parallels Maslov's more basic list of human needs. Customer satisfaction increases as the product fulfills requirements higher up in this hierarchy. This understanding gives a design team more information for determining priorities on customer requirements. For example, Expecters must be satisfied first because they are the basic characteristics that a product is expected to possess. Customer complaints tend to be about expecter-type requirements. Therefore, a product development strategy aimed solely at eliminating complaints may not result in highly satisfied customers.

Spokens give greater satisfaction because they go beyond the basic level and respond to specific customer desires. Unspokens are an elusive category that the team must capture through indirect methods like ethnographic research. True Exciters will serve to make a product unique. Often companies will introduce innovative technology into a product expecting it to become an Exciter in Kano terms.

Considering all the information on customer requirements that has been presented up to this point, the design team can now create a more accurately prioritized list of customer requirements. This set is comprised of

- Basic CRs that are discovered by studying competitor products during benchmarking
- Unspoken CRs that are observed by ethnographic observation
- High-ranking customer requirements (CRs) found from the surveys
- Exciter or Delighter CRs that the company is planning to address with new technology.

The highest-ranked CRs are called *critical to quality customer requirements* (CTQ CRs). The designation of CTQ CRs means that these customer requirements will be the focus of design team efforts because they will lead to the biggest payoff in customer satisfaction.

3.4 ESTABLISHING THE ENGINEERING CHARACTERISTICS

Establishing the engineering characteristics is a critical step toward writing the product design specification (Sec. 3.6). The process of identifying the needs that a product must fill is a complicated undertaking. Earlier sections of this chapter focused